LINUX SECURITY BASICS

Task 19 — Internship Practical Report

Internship Track	Cyber Security
Organization / Platform	Veda Technology Internship Dashboard
Practical Environment	JSLinux online emulator (Alpine Linux, x86_64 console)
Date performed	19 October 2026
Working directory	linux_security
Test file	practical.txt

1. Aim

To practise basic Linux security controls by creating a file, viewing its permissions and ownership, changing permission modes with **chmod**, and checking the current user and group information.

2. Objectives

• Understand read, write and execute permissions.

• Use chmod to change file permissions.

• View file metadata and ownership with ls -l and stat.

• Identify current user and group information using whoami, id and groups.

• Relate file permissions to the principle of least privilege.

3. Tools and Environment

A web browser and JSLinux were used to access an Alpine Linux x86_64 console. The practical was completed in an online emulator, so no Linux installation on the Windows laptop was required.

4. Procedure and Commands

Create a working directory and file

```
mkdir linux_security
cd linux_security
touch practical.txt
```

View file details

```
ls -l practical.txt
```

Set owner read/write permissions

```
chmod 600 practical.txt
ls -l practical.txt
```

Set owner read/write and group/others read-only

```
chmod 644 practical.txt
ls -l practical.txt
```

Give owner all permissions only

```
chmod 700 practical.txt
ls -l practical.txt
```

Check account and file metadata

```
whoami
id
groups
stat practical.txt
```

5. Results and Observations

Command / mode	Observed meaning
chmod 600	Output -rw-------: owner can read and write; group and others have no permissions.
chmod 644	Output -rw-r--r--: owner can read/write; group and others can read.
chmod 700	Output -rwx------: owner can read, write and execute; group and others have no permissions.
whoami	Output was root.
id	Output showed uid=0(root) and gid=0(root).
groups	Output showed root.
stat practical.txt	Confirmed file metadata, permission mode 0700 and root ownership.

6. Security Concepts Learned

• **chmod:** Changes file and directory permissions.

• **File ownership:** Identifies the user and group associated with a file.

• **Least privilege:** Grant only the access needed for a task.

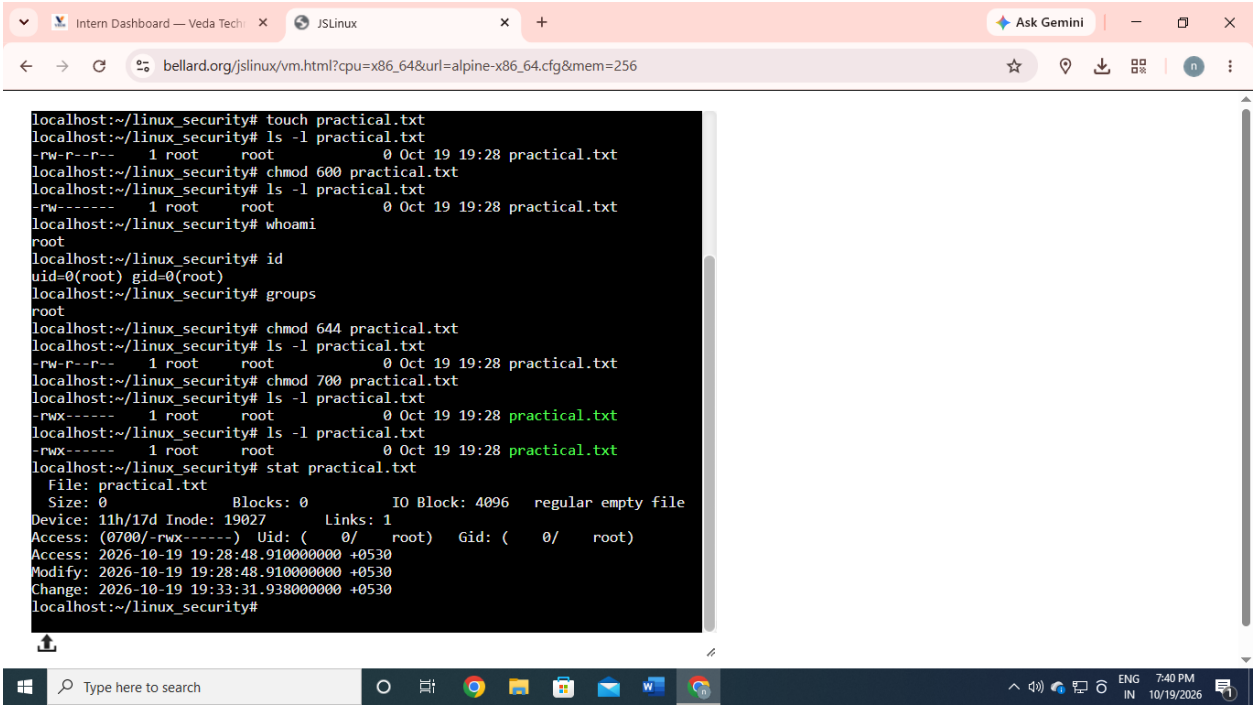
• **Permission notation:** r = read, w = write, x = execute; permissions are defined for owner, group and others.

7. Conclusion

The practical demonstrated file creation, permission inspection and permission changes using chmod. The whoami, id, groups and stat commands were used to inspect account and file information. The emulator session ran as root, so this demonstrates command usage but not operation from a least-privileged standard user account.

8. Practical Evidence — Updated Screenshot

The following screenshot was supplied after the practical was completed. It replaces the previous screenshot in this report and shows the terminal commands, permission modes 600, 644 and 700, and the stat output.


Intern Dashboard — Veda Tech
JSLinux
Ask Gemini
bellard.org/jslinux/vm.html?cpu=x86_64&url=alpine-x86_64.cfg&mem=256
localhost:~/linux_security# touch practical.txt
localhost:~/linux_security# ls -l practical.txt
-rw-r--r--   1 root   root   0 Oct 19 19:28 practical.txt
localhost:~/linux_security# chmod 600 practical.txt
localhost:~/linux_security# ls -l practical.txt
-rw-------   1 root   root   0 Oct 19 19:28 practical.txt
localhost:~/linux_security# whoami
root
localhost:~/linux_security# id
uid=0(root) gid=0(root)
localhost:~/linux_security# groups
root
localhost:~/linux_security# chmod 644 practical.txt
localhost:~/linux_security# ls -l practical.txt
-rw-r--r--   1 root   root   0 Oct 19 19:28 practical.txt
localhost:~/linux_security# chmod 700 practical.txt
localhost:~/linux_security# ls -l practical.txt
-rwx------   1 root   root   0 Oct 19 19:28 practical.txt
localhost:~/linux_security# ls -l practical.txt
-rwx------   1 root   root   0 Oct 19 19:28 practical.txt
localhost:~/linux_security# stat practical.txt
  File: practical.txt
  Size: 0   Blocks: 0   IO Block: 4096   regular empty file
Device: 11h/17d Inode: 19027   Links: 1
Access: (0700/-rwx------) Uid: (0/   root)   Gid: (0/   root)
Access: 2026-10-19 19:28:48.910000000 +0530
Modify: 2026-10-19 19:28:48.910000000 +0530
Change: 2026-10-19 19:33:31.938000000 +0530
localhost:~/linux_security#
Type here to search
ENG
IN
7:40 PM
10/19/2026